

Smart Plant Carer · Premium Course

2-session series (2 hrs each) · 9–13 yrs · from KES 7,000

Premium course outline **PREMIUM**

Overview

Build a self-watering planter that senses dry soil and pumps water to a real seedling all by itself.

DETAIL

Track	Arduino Robotics & Electronics
Format	2-session series (2 hrs each)
Ages	9–13 yrs
Level	Beginner → Intermediate
Price	from KES 7,000
Maker Kit	Optional, KES 8,500 one-time: Arduino Uno R3 board + USB cable, Breadboard + 40 jumper wires, LEDs (red, yellow, green, white, RGB) + resistors, Push buttons, potentiometer and piezo buzzer, Light sensor (LDR) and sound sensor, HC-SR04 ultrasonic sensor + bracket, IR sensor, DHT11 temperature & humidity sensor, PIR motion sensor, SG90 servo motor, Relay module, 16×2 LCD screen (I2C), Battery holder, Storage box. Bought once and reused in every Arduino course. Kits are lent during sessions; buy one to keep building at home.
Parts pack	Optional, KES 1,000: Soil-moisture sensor, Mini water pump + tubing + reservoir, Pot and seedling
Standalone	Yes
Prerequisites	None — standalone

Programme

1. **Sense & pump.** Wire a soil-moisture sensor and calibrate the dry vs. wet readings on the Serial Monitor, then drive a small water pump through a relay module.

2. **Automatic planter.** Set a moisture threshold so it waters only when the soil is dry, then assemble the pump, tubing and reservoir around a potted seedling and run a full automatic watering cycle.

What they take home

 Self-watering planter kit + a seedling

Why parents love it

“They build tech that keeps a plant alive on its own — and understand every part of why it works.”

How to book

Places are limited and run on scheduled dates. To book or ask a question, email **tinkerwithanne@gmail.com** with the course name (**Smart Plant Carer**), your child's age, and your preferred dates. Prices start from the amount above and may vary with group size and materials.